

Poetic Personality Chatbot using Google Gemini

Project Process Report

Introduction

This project develops an AI-powered chatbot named 'Luna – The Eternal Poet' that converts ordinary user messages into expressive poetic verses using Google's Gemini Large Language Model. The objective is to demonstrate Prompt Engineering, conversational AI, and Generative AI.

Steps Followed

1. Understood LLM concepts and project objectives.
2. Configured the Google Gemini API and Python SDK.
3. Built reusable prompt templates.
4. Designed Luna's poetic personality.
5. Implemented poetry generation functions.
6. Developed an interactive chatbot.
7. Added typing animation, conversation memory, and emotion-aware prompting.
8. Evaluated the chatbot using qualitative test cases.

Technologies Used

Python, Google Gemini API (google-genai SDK), Jupyter Notebook, Prompt Engineering, Object-Oriented Programming.

Challenges

API authentication, SDK migration, free-tier quota limitations, prompt refinement, maintaining a consistent personality, and handling runtime exceptions.

Solutions

Updated to the latest SDK, implemented modular functions and chatbot classes, improved prompts iteratively, added conversation memory, and implemented graceful exception handling.

Conclusion

The project demonstrates how Prompt Engineering can transform a general-purpose LLM into a creative conversational agent capable of generating emotionally expressive poetry while maintaining a consistent personality.

Reflection

Prompt engineering shaped the chatbot's creative identity by defining its personality, tone, and response style. Through iterative testing, I learned that carefully designed prompts often have a greater impact on output quality than code complexity. The project also provided practical experience with API integration, conversational AI, modular software development, and handling real-world limitations such as API quotas and runtime errors.